

Chapter 62 — Paediatric Genitourinary Trauma

RENAL INJURIES

Epidemiology

- **The kidney is the commonest site of paediatric genitourinary (GU) trauma.**
- **The child's kidney is thought more vulnerable** (table below); a more pliable spine also allows more ureteric stretch injury. **Whether children are in fact more susceptible remains unproven.**
- **A minority of children with blunt trauma — roughly 5–20% — have a renal injury.**
- National trauma registry data (United States and Japan): **about one-quarter of all renal injuries occur before 19 years of age**; these are **overwhelmingly blunt** and **mostly grade 1–3**. Mean age is 14 years; children under 5 years, and especially infants, are a small minority.
- **Open surgery rises steeply with grade**: a minority of grade 4 and about half of grade 5 injuries underwent open operation in United States data; in Japanese data nephrectomy occurred only in grade 4–5, never in grade 1–3.
- **Minimally invasive treatment** (angioembolisation, ureteral stent) is uncommon overall and more frequent in grade 4–5; angiography is used more liberally in Japan than in the United States.
- **Nephrectomy is rare — about 3–5% of admissions — but more likely when children are managed in adult rather than paediatric trauma units.**
- **Causes**: motor vehicle and motorcycle collisions, falls, bicycle accidents and sport; **all-terrain vehicle injuries may be increasing.**
- **Congenital anomalies** such as hydronephrosis and ureteropelvic junction (UPJ) obstruction are found **three to five times more often in children screened by computed tomography (CT) for trauma than in adults**. Whether an anomaly raises the risk of injury is unclear; **injured anomalous kidneys usually sustain low-grade injury.**

Presentation

- **Blunt trauma predominates**: motor vehicle collisions, falls, sport, pedestrian versus vehicle, all-terrain vehicle and motorcycle collisions.
- **Other organ injuries accompany about half of blunt renal trauma** — liver, spleen, head, orthopaedic.
- **Rapid deceleration** (road collision, fall from height) is the historical feature associated with renal injury.
- **Examination**: flank tenderness or ecchymosis in severe injury; rib fractures; tachycardia and hypotension with major blood loss.
- **Hypotension is late and unreliable in children** (table below) — **a widely cited paradigm resting on limited data**. **Absence of haematuria does not exclude significant injury.**
- **The urologist is usually consulted after the diagnosis is made**; the decision to image lies with emergency and trauma physicians.

CHILD VERSUS ADULT: RENAL TRAUMA

Feature	Child	Adult
Renal protection	Less perirenal fat, weaker abdominal wall, kidney below rib cage	Better cushioned and covered
Blood pressure after haemorrhage	Maintained until late (thought 25–30% volume loss)	Falls earlier; a more reliable sign
Haematuria	May be absent with significant injury	Classic imaging trigger

Feature	Child	Adult
Imaging decision	Mechanism-led	Haematuria- and hypotension-led
Radiation	Cancer risk greater the younger the child — ALARA	Less weight given

Concept — *The child's kidney is anatomically exposed and the child's circulation compensates silently, so neither blood pressure nor urinalysis can be trusted to flag injury; mechanism must.*

INDICATIONS FOR RENAL IMAGING

- **Adult triggers: gross haematuria, or microscopic haematuria with hypotension (systolic blood pressure < 90 mm Hg).** Highly sensitive for blunt and penetrating injury but **poorly specific** — most patients meeting them have no identifiable GU injury.
- **In children both triggers are weaker:** up to two-thirds with a grade ≥ 2 injury may have a normal urinalysis; hypotension is late.
- **Mechanism is therefore a central predictor in children:** significant deceleration or high-velocity injury — **fall from more than 10 feet (about 3 m)**, high-speed motor vehicle collision, or a blow to abdomen or flank from a foreign object.
- **Associated injuries that predict GU injury:** thoracic or intra-abdominal organ injury; fractures of **ribs, spine, femur or pelvis**.
- **Four criteria in children — haematuria, hypotension, associated injuries, mechanism — give more than 98% sensitivity** for significant GU injury.
- **Penetrating trauma: every abdominal gunshot wound** requires GU evaluation; **stab wounds to the flank, upper abdomen or lower chest** require renal imaging.
- **American Urological Association (AUA) Urotrauma Guideline and European Association of Urology (EAU) Guidelines on Urological Trauma** endorse the criteria in Table 62.1, adding associated abdominal injury.

TABLE 62.1 INDICATIONS FOR IMAGING FOR RENAL TRAUMA

Setting	Indication
Stable blunt trauma	Gross haematuria
Stable blunt trauma	Microscopic haematuria and systolic blood pressure < 90 mm Hg
Mechanism or examination	Rapid deceleration; significant blow to flank; rib fracture; flank ecchymosis
Penetrating	Injury to abdomen, flank or lower chest

Modified from Morey AF et al., Urotrauma: AUA guideline, J Urol 2014.

- **What the table tests:** haematuria is only one of four routes into imaging. **Penetrating injury qualifies with no haematuria and a normal blood pressure.** Trap: "no haematuria, so no imaging".

Concept — *Adult criteria are haematuria-led; paediatric criteria are mechanism-led, because the two adult markers fail precisely in children.*

IMAGING STUDIES FOR DIAGNOSIS

As Low as Reasonably Achievable

- CT for blunt abdominal trauma is ubiquitous and **probably overused for mild trauma in children**.
- CT radiation carries a **very small absolute increase in later cancer (leukaemia, brain tumours), greater the younger the child at scanning** → **ALARA (as low as reasonably achievable)** must govern both diagnostic and follow-up imaging.

Focused Assessment With Sonography for Trauma

- **Focused assessment with sonography for trauma (FAST):** sensitivity for renal injury highly variable and often low (reported range 22–96%).
- **Why:** performed by emergency clinicians to look for free fluid in dependent regions, not to examine organs.
- A normal FAST plus normal serial physical examination over 24 hours reliably excludes clinically significant renal or intra-abdominal injury.
- None of the studies was paediatric-specific.

Dedicated Renal Ultrasonography

- Little studied against CT in children or adults.
- Bottom line in stable children: missed injuries have been grade 1 and insignificant; focused and colour Doppler studies reliably detected grade 3–5, and every grade 3–5 injury had gross haematuria or ≥ 50 red blood cells per high-power field (RBC/hpf).
- Contrast-enhanced ultrasonography is used by some centres for diagnosis and follow-up; prospective comparative studies are lacking.
- AUA: ultrasound may be used initially, CT preferred; EAU Guidelines on Paediatric Urology: ultrasound to screen, CT to diagnose and stage.

IMAGING MODALITIES FOR SUSPECTED RENAL INJURY

Modality	Radiation	Strength	Weakness	Guideline role
FAST	None	Free fluid; + 24 h serial examination excludes significant injury	Does not assess the kidney	Not a staging test
Dedicated renal ultrasound	None	Detects grade 3–5	Few studies; misses grade 1	Initial (AUA) / screening (EAU Paediatric)
Contrast-enhanced ultrasound	None	Some centres	No prospective data	—
Contrast CT, two-phase	Yes	Most sensitive and specific	Radiation	Preferred

Concept — Ultrasound's weakness is purpose rather than physics: **FAST looks for fluid, a dedicated renal study looks at the kidney** — and the latter may one day replace CT in stable low-grade injury.

COMPUTED TOMOGRAPHY

- In the haemodynamically stable patient, CT is the most sensitive and specific test to diagnose and exclude GU trauma.
- AUA and EAU: intravenous (IV) contrast-enhanced CT of abdomen and pelvis with immediate and delayed phases.
- Children often receive a single contrast phase — to limit radiation, or from unfamiliarity with the recommendation.
- Cost of a single phase: reduced accuracy for urinary extravasation and isolated ureteral injury, especially with a urinoma or deep or multiple lacerations.
- Counterpoint: in the era of non-operative management, whether a routine delayed phase changes outcome in children is uncertain.

- **Practical solution:** the radiologist reviews the nephrographic phase in real time and **adds a delayed phase only if a urine leak is suspected** (perinephric collection, deep or multiple lacerations). The EAU accepts that the initial scan is often single-phase, with further phases when the injury is not well characterised.

Concept — The delayed phase exists to find urine, not blood; add it selectively when the nephrographic phase raises the possibility of a leak.

CLASSIFICATION AND GRADING OF RENAL INJURIES

The American Association for the Surgery of Trauma Grading System

- The American Association for the Surgery of Trauma (AAST) renal Organ Injury Scale (OIS), **1989, is the most widely used**; a **proposed revision in 2011** aimed to clarify high-grade (4–5) injuries.
- **Grade predicts intervention and complications** in adults and children.
- Because intervention, complications and loss of function are negligible below grade 4, **grade 1–3 = low grade, grade 4–5 = high grade**.

TABLE 62.2 RENAL INJURY: 1989 AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SCALE AND 2011 REVISED INJURY STAGING CLASSIFICATION

Grade	1989 AAST OIS	2011 proposed revision
1	Non-expanding subcapsular haematoma, no laceration; micro- or gross haematuria with normal urological studies	Subcapsular haematoma and/or contusion; no collecting system injury
2	Non-expanding perirenal haematoma confined to renal retroperitoneum; cortical laceration < 1 cm, no extravasation	Laceration < 1 cm into cortex; small haematoma within Gerota fascia; no collecting system injury
3	Laceration > 1 cm into parenchyma; no collecting system rupture or extravasation	Laceration > 1 cm into medulla ; haematoma within Gerota fascia; no collecting system injury
4	Laceration through cortex, medulla and collecting system; main renal artery or vein injury with contained haemorrhage	Laceration(s) into collecting system with urinary extravasation; segmental artery or vein injury; renal pelvis laceration and/or complete UPJ disruption
5	Completely shattered kidney; avulsion of renal hilum, devascularising the kidney	Main renal artery or vein laceration, avulsion or thrombosis

Modified from Moore EE et al., J Trauma 1989; Buckley JC, McAninch JW, J Trauma 2011.

- **Pattern:** grades 1–3 differ only in depth (< 1 cm versus > 1 cm) and **never involve the collecting system; entry into the collecting system is grade 4**.
- **Traps:** the shattered kidney is **grade 5 in 1989 but grade 4 in 2011**; main-vessel injury with contained haemorrhage is **grade 4 in 1989**, whereas in 2011 any main-vessel laceration, avulsion or thrombosis is **grade 5**; UPJ disruption, unlisted in 1989, is **grade 4 in 2011**.

Figure Legends

- **Fig. 62.1** Grade 2 right renal laceration with perirenal haematoma on immediate and delayed phases; incidental **bifid collecting system** with separate upper-pole ureter.
- **Figs. 62.2 and 62.3** Grade 3 (> 1 cm, no delayed-phase extravasation) and grade 4 (**delayed-phase extravasation from the collecting system**) lacerations.

High-Risk Criteria

- Within grade 4–5, specific CT findings predict intervention:
 - **Medial contrast extravasation** → possible UPJ disruption.
 - **No contrast in the ureter with a grade 4 injury** → later intervention for urine leak (paediatric series).

- **More than 25% of parenchyma devitalised** → significantly more angioembolisation and open surgery.
- **Bleeding predictors: perirenal haematoma > 3.5 cm (in children > 2.2 cm), vascular contrast extravasation, medial laceration.**
- **Risk is additive: 0–1 bleeding risk factor → < 10% intervention; 2–3 → > 50%.**
- The criteria hold in children, and many advocate building them into the AAST scale (substratification of grade 4 into low-risk 4a and high-risk 4b).

TABLE 62.3 HIGH-RISK CRITERIA FOR INTERVENTION IN GRADES 4 TO 5 RENAL INJURIES

Predicts	Criterion
Bleeding (angioembolisation)	Large perirenal haematoma: > 3.5 cm adults, > 2.2 cm children
Bleeding	Vascular contrast extravasation
Bleeding	Medial renal laceration
Bleeding and open surgery	> 25% of renal parenchyma devitalised
Urine leak / UPJ disruption	Medial extravasation of urine on excretory phase
Urine leak needing intervention	Urine extravasation with no contrast in the distal ureter on excretory phase

Modified from Au 2016; Canon 2014; Dugi 2010; Figler 2013; Hardee 2013; Long 2013.

- **What the table is for:** splitting grade 4 into a group that behaves like low-grade injury and a group likely to need a procedure. **The medial findings (laceration, urine) point to the hilum and UPJ; the haematoma and vascular findings point to bleeding.** Trap: the paediatric haematoma threshold is smaller (2.2 cm).

Blunt Versus Penetrating

- The distinction determines imaging and exploration.
- **Any significant penetrating injury to abdomen, flank or lower chest → GU evaluation, even if stable and without haematuria.**
- **Unstable, or laparotomy needed for other injuries → inspect retroperitoneum and ureters, with high suspicion for renal or ureteral injury, especially with high-velocity gunshot wounds.**

BLUNT VERSUS PENETRATING RENAL TRAUMA

	Blunt	Penetrating
Frequency in children	Great majority	Uncommon
Imaging trigger	Haematuria, hypotension, mechanism, associated injury	Injury location alone (abdomen, flank, lower chest)
Haematuria required	No, but a key criterion	No
Non-operative management	Default at every grade	Favourable outcomes in stable patients
At laparotomy	Retroperitoneum inspected if unstable	Inspect retroperitoneum and ureters; high-velocity wounds most dangerous

Management

- **Most paediatric renal trauma, including high-grade injury, is managed non-operatively; initial non-operative management is a strong recommendation** of the EAU paediatric and Pediatric Trauma Society guidelines.
- **Grade 1–3: the vast majority need no intervention, let alone open surgery.**
- **Grade 4–5: higher risk, but most still need no intervention.** Interventions are usually **endoscopic or percutaneous** — percutaneous nephrostomy, ureteral stent, percutaneous drain, angioembolisation.

Nonoperative

- Numerous adult and several paediatric series confirm **safety and efficacy**. **Nephrectomy rates are higher with surgical exploration than with non-operative management**, in adults and children.
- **Eligibility**: haemodynamically stable, or **resuscitated adequately and remaining stable**.
- **Ideal candidate**: grade 1–3 blunt injury, no associated injury requiring surgery, haemodynamically stable.
- **Grade 4–5 injuries have been managed non-operatively with success** in several centres; **non-operative management of penetrating renal trauma also has favourable outcomes**.

Nonoperative Management Protocol

- **Sequence**: confirm haemodynamic stability → grade the injury → bed rest → serial haemoglobin/haematocrit → serial abdominal examination → **bed rest until gross haematuria resolves and blood counts are stable**.
- **Grading imaging**: AUA and EAU state dedicated renal ultrasonography is an option in children, but **contrast CT abdomen and pelvis with immediate and delayed phases is preferred**. In future, ultrasound may suffice to exclude significant retroperitoneal haematoma, deep laceration or urinoma and so **avoid CT in stable grade 1–3 injury**.
- **Intensity scales with severity**:
 - **Grade 1–2, no gross haematuria, stable haemoglobin** → 23-hour observation with two blood tests.
 - **Grade 4 with large perirenal haematoma, vascular extravasation and ongoing gross haematuria** → intensive care, bloods every 4–6 hours, possible transfusion and angioembolisation.

Early Repeat Imaging

- **Guidelines diverge** (table below): AUA and EAU Urological Trauma favour repeat CT for high grade; EAU Paediatric favours ultrasound first.
- **Adults** (two large studies and a systematic review): early repeat imaging **may be unnecessary even in high grade, if asymptomatic and progressing well**.
- **Children with grade 3–5 injury**: **no difference in complications, delayed procedures or readmissions with or without early repeat imaging; symptoms, not imaging, predicted complications and delayed procedures**.
- **Repeat CT offers no clear advantage over ultrasound in a stable child**. Authors: **ultrasound at 48–72 hours for high grade**; CT only if ultrasound is inadequate or symptoms concern. Even routine 48-hour ultrasound in well grade 4–5 is questioned.

GUIDELINE POSITIONS: PAEDIATRIC RENAL TRAUMA

Decision	AUA Urotrauma	EAU Urological Trauma	EAU Paediatric Urology	Pediatric Trauma Society
Imaging triggers	Gross haematuria; micro + SBP < 90; deceleration; flank blow; suspicious findings	Same	—	—
Initial imaging in children	Ultrasound may be used; CT preferred	CT immediate + delayed	Ultrasound to screen; CT best to diagnose and stage	—
CT phases	Immediate + delayed	Immediate + delayed; single phase acceptable, add phases if needed	—	—
Initial management	Non-operative	Non-operative	Non-operative (strong)	Non-operative (strong)

Decision	AUA Urotrauma	EAU Urological Trauma	EAU Paediatric Urology	Pediatric Trauma Society
Early repeat imaging	CT at 48 h, grade 4–5	CT at 2–4 days, high-grade and penetrating	Ultrasound at 2–3 days, grade ≥ 2 , before CT	—
Urine leak	Non-operative initially; stent first (expert opinion), add drain	Non-operative initially; stent and/or drain	—	—
Follow-up imaging	None for grade 1–2	Individualised	—	—
Blood pressure	—	—	—	Check at follow-up; interval undefined

(SBP = systolic blood pressure, mm Hg.)

Outcomes of Nonoperative Management

- **Lower nephrectomy rate with non-operative than operative management**, in adult and paediatric series.
- **The vast majority of children are managed non-operatively.** Intervention risk is **significantly higher in grade 4–5 and penetrating injuries**, but **most stable patients succeed non-operatively**.

Management of Complications in Nonoperative Patients

- **Complication risk rises with grade.**
- **Grade 1–2, and grade 3 without devitalised segments: complication risk close to zero.**
- **Grade 3 with devitalised segments to grade 5 (stable patients):** about **25% bleeding requiring angioembolisation**, about **15% urinoma requiring intervention**, and only about **5% open exploration** for a complication beyond percutaneous or endoscopic rescue.

Persistent Urine Leak

- **Grade 1–3 by definition do not enter the collecting system; grade 4–5 can leak.**
- **Commonest leak site: the parenchymal laceration;** also renal pelvis rupture, ureteral injury or UPJ disruption.
- **AUA and EAU: renal injury with urine leak is managed non-operatively initially.**
- **Always evaluate the renal pelvis and ureter for UPJ disruption or pelvic rupture** — delayed-phase CT or retrograde pyelography.
- **Up to 85% of urine leaks resolve spontaneously;** the rest need percutaneous urinoma drainage or diversion by nephrostomy and/or ureteral stent.
- **Foley catheter** is often placed for 2–3 days to drain the bladder — benefit for speed or likelihood of resolution unproven.
- **IV antibiotics (typically cefazolin)** are common with a leak — **no evidence that they reduce infection;** practice varies.

Indications for Intervention for Urine Leak or Urinoma

- **Classic triad of symptomatic persistent urinoma: ipsilateral flank pain, ileus, low-grade fever.**
- **Shared decision-making with the family rather than a strict algorithm:** very large, symptomatic urinoma → early intervention; small, mildly symptomatic urinoma → longer observation, which may avoid intervention.

- **Urine leak with no contrast in the distal ureter on the urographic phase** → possible UPJ disruption or major pelvic injury needing immediate repair or at least drainage → **cystoscopy with retrograde pyelography**.
- If retrograde pyelography excludes UPJ disruption and pelvic laceration, **some still place a ureteral stent**, because **leak plus absent ureteric contrast on initial CT predicts symptomatic urinoma without a stent**.
- **Fig. 62.5** UPJ disruption with a devitalised lower pole: medial extravasation, empty ureter on intravenous pyelography, **confirmed by retrograde pyelography**.

INTERVENTIONS FOR SYMPTOMATIC URINOMA OR PERSISTENT LEAK

Intervention	Place	Advantages	Disadvantages
Ureteral stent (+ Foley)	First line (AUA, expert opinion) ; resolves about 90% of symptomatic urinomas	Minimally invasive; no external tube to dislodge	Anaesthesia to place and remove ; bothersome stent symptoms, especially in pubertal children; blockage; harder to confirm resolution
Percutaneous drain	After failed stent, especially a very large urinoma; may be first line	Rapid symptom relief ; output can be monitored; removed without anaesthesia	Anaesthesia and interventional radiology to place; dislodgement; may still need a stent
Nephrostomy tube	Rarely alone ; with antegrade stent after failed stent + drain, or when lithotomy is impossible	High resolution with antegrade stent	Renal injury and bleeding risk
Angioembolisation of isolated segment	Leak from a segment not in continuity with the collecting system , failing all the above	Avoids open surgery	Very rare; limited reports
Open repair	After prolonged observation and multiple failed minimally invasive procedures	Definitive	Risk of renal loss or nephrectomy

Ureteral Stent

- **Authors' sequence**: Foley kept **1–2 days after urinoma symptoms resolve** → ultrasound for urinoma size → remove Foley → **observe at least one further day** for recurrence → **stent removed at 4–6 weeks**, considering ultrasound first.
- IV cefazolin while the Foley is in place, and prophylactic antibiotics while the stent is in place, are common but **unsupported by good evidence**.

Percutaneous Drain

- **Fig. 62.6** Grade 4 laceration, fever and pain at day 2 with persistent extravasation → **drain alone, no Foley** → **resolved in 3 days**, removed without anaesthesia.
- **Sequence**: drain (discuss a Foley as an adjunct) → follow output → **remove when output is minimal and ultrasound shows resolution**. Ultrasound after symptoms settle is of unclear necessity but usual.

Nephrostomy Tube

- **Rarely used alone** — **stent or drain preferred first because nephrostomy risks renal injury and bleeding**; with an antegrade stent it **resolves most leaks**, and it suits the child who **cannot be placed in lithotomy**.

Angioembolization of Isolated Renal Segment

- **Segment not in continuity with the collecting system, leaking despite stent, nephrostomy and drain → embolise it. Very rare.**

Renal Exploration With Repair of Urine Leak

- **Rare; only after prolonged observation and failed minimally invasive procedures. Sequence:** debride devitalised tissue → close collecting system with absorbable suture → repair parenchyma over a bolster → drain with ureteral stent and perinephric drain. **Counsel about renal loss or nephrectomy.**

Concept — Non-operative management is the default at every grade; grade and the high-risk CT criteria set only the intensity of observation and the likelihood of a minimally invasive rescue. Urine is diverted from the least invasive route outward — stent, drain, nephrostomy, embolisation of an orphan segment, open repair last — and absent distal ureteric contrast is the one finding that forces early retrograde study.

PERSISTENT OR DELAYED BLEEDING

Indications for Angioembolization

- **About 25% of grade 3–5 injuries managed non-operatively develop persistent or delayed bleeding needing intervention.** Predictors are the high-risk criteria: haematoma size, vascular extravasation, medial laceration.

PERSISTENT VERSUS DELAYED RENAL BLEEDING

	Persistent bleeding	Delayed bleeding
Mechanism	Injured vessels fail to tamponade	Arteriovenous fistula or pseudoaneurysm ruptures; possibly as the tamponading haematoma liquefies or resolves
Timing	From injury	Typically 1–2 weeks; up to 1 month
Stable patient	Allow 2–3 transfusions before angioembolisation	Early angioembolisation — most delayed bleeds need intervention — unless mild and self-limited
Unstable or rapid bleeding	Urgent angioembolisation	Urgent angioembolisation

- **Superselective angioembolisation succeeds in more than 80% of persistent and delayed bleeds, in adults and children; a failed first attempt can be repeated successfully.**
- Performed by interventional radiology; **aim: embolise the least viable parenchyma that stops the bleeding.** Some functional loss is inevitable, depending on the injury and the procedure.
- **Retained differential function: grade 4 about 35–45%; grade 5 about 10–30%.**
- **Complications:**
 - **Postembolisation syndrome** — self-limiting flank pain, fever, possible ileus; < 10%; resolves in 3–4 days.
 - **Persistent fever → suspect postembolisation abscess → repeat CT → drain any concerning collection.**
 - **Persistent or recurrent bleeding in a minority → repeat embolisation if stable → exploration if bleeding persists after repeat embolisation.**

Renal Exploration

- Needed in a **small percentage** of non-operatively managed patients with persistent or delayed bleeding; these patients are usually **more stable** than those explored at presentation.
- **Goals, in order:** control renal vasculature → expose the whole injury → debride devitalised tissue → close collecting system defects → close parenchyma over a bolster → drain.

KEY POINTS: NONOPERATIVE MANAGEMENT AND COMPLICATIONS DURING NONOPERATIVE MANAGEMENT

- Non-operative management for most, including grade 4–5 (higher intervention risk).
- Bleeding in about 25% of grade 3–5 → **superselective angioembolisation, repeatable**.
- About 15% of urine leaks need a stent and/or drain.
- Repeat CT at 48–72 h: **not for well grade 1–3; optional, of unclear value, in well grade 4–5**.

Concept — Renal bleeding is solved endovascularly: superselective embolisation is first line for persistent and delayed haemorrhage alike and is repeated before surgery is contemplated — **its price is a measured loss of parenchyma, larger the higher the grade**.

OPERATIVE

Indications

- **Primary indication:** persistent haemodynamic instability from renal bleeding.
- Pulsatile or expanding retroperitoneal haematoma at laparotomy (no prior imaging) → explore the kidney.
- Renal bleeding uncontrolled by selective angioembolisation.
- **Stable, non-pulsatile haematoma** (inadequately staged, or known devitalised segments): once explored after penetrating trauma; **now left alone, even after gunshot wounds** — nephrectomy rates may be lower, and high-grade paediatric injury tolerates non-operative care. **EAU agrees**; if another organ obviously explains the instability, **avoid renal exploration**.

RETROPERITONEAL HAEMATOMA FOUND AT LAPAROTOMY

Pulsatile or expanding	Stable, non-pulsatile
Explore — evidence of ongoing renal bleeding causing instability	Leave alone (EAU) — even with gunshot wounds; exploration may raise nephrectomy rate

One-Shot Intraoperative Intravenous Pyelography or Ultrasonography

- **Purpose:** confirm a contralateral kidney when preoperative imaging has not done so.
- **Technique:** 2 mL/kg IV contrast bolus → single radiograph at 10–15 minutes.
- It shows **only whether a contralateral kidney is present**; it **does not grade the injury**. It informs nephrectomy versus reconstruction.
- **Intraoperative Doppler ultrasonography** is an alternative (presence and perfusion of the contralateral kidney) but less reported and less recommended.

Control of Renal Vasculature

- **Early vascular control:** incision in the posterior mesentery medial to the inferior mesenteric vein, over the aorta → vessel loops around renal vessels, tightened if needed.
- Retrospective series suggested early control lowered nephrectomy rates; **the one randomised trial found no difference in nephrectomy and longer operating time with early control**.
- **Sensible in the unstable patient with an expanding haematoma**; AUA and EAU guidelines do not address it.

Reconstruction Versus Nephrectomy

- **Goals:** stop life-threatening haemorrhage and salvage the kidney whenever feasible and safe (EAU).

- **Sequence:** confirm contralateral kidney → open Gerota fascia → control main vessels → expose entire injury → debride → oversee bleeding vessels and close collecting system (absorbable) → **small defects: capsular sutures; large: close over haemostatic bolster or omentum** (authors: sliding-clip renorrhaphy).
- **Nephrectomy if:** the injury is irreparable or would leave a very small remnant; or **the patient is unstable** — hypothermia, coagulopathy, ongoing blood loss — and speed saves life.
- **Penetrating (especially high-velocity gunshot) injury:** inspect and repair the ipsilateral ureter.

Renal Arterial Laceration

- The kidney lacks good collateral arterial flow, so segments devitalised by arterial injury are not salvageable.
- **Avulsed or significantly bleeding main renal artery → unstable → explore expecting nephrectomy; repair attempts carry poorer outcomes.**
- **Exception: solitary kidney or bilateral arterial injury → attempt repair**, with vascular surgery if available.

Renal Arterial Occlusion

- Complete or partial occlusion after blunt trauma from **renal artery embolism or an intimal flap**.
- **Observation has succeeded;** expect **post-infarction syndrome (fever, flank pain)** and **follow for hypertension**.
- **Nephrectomy** for uncontrolled pain or infection during observation.
- **Percutaneous revascularisation (angioplasty and/or endovascular stent)** has salvaged some completely or partially occluded kidneys.

RENAL ARTERIAL LACERATION VERSUS OCCLUSION

	Laceration or avulsion	Occlusion (embolism, intimal flap)
Presentation	Haemodynamically unstable	Stable; infarcted or ischaemic kidney
Default	Explore; anticipate nephrectomy	Observe; endovascular revascularisation an option
Repair	Worse outcomes — only for solitary kidney or bilateral injury	Angioplasty ± stent
Sequelae to watch	—	Post-infarction syndrome; hypertension

Ureteropelvic Junction Disruption

- **Rare but important not to miss.**
- **Mechanism: high-velocity acceleration-deceleration and torso hyperextension** — the mobile kidney is displaced away from the fixed proximal ureter.
- **Associated injuries: Chance fracture** (flexion-distraction injury: anterior vertebral body compression with a transverse fracture through the posterior vertebra) and other severe injuries, such as **aorta and duodenum**.
- Associated injuries often prompt **laparotomy without preoperative imaging; haematuria is usually present but absent in a significant minority → diagnosis is often delayed → retroperitoneal abscess, fistula, infected urinoma**.
- **Delayed clues:** persistent postoperative fever, ileus, flank pain, **urine in intra-abdominal drains**, enlarging urinoma → repeat CT.
- **Classic CT findings:** medial extravasation near the UPJ, no contrast in the distal ureter, no parenchymal laceration. A laceration does not exclude UPJ disruption. Any suspicion → retrograde pyelography.

- **Fig. 62.7** Bilateral UPJ disruption: minimal lacerations, **bilateral medial delayed-phase extravasation, Chance-type fracture T12–L1.**

UPJ DISRUPTION: IMMEDIATE VERSUS DELAYED REPAIR

Immediate repair	Nephrostomy + drain, delayed repair
Stable patient; diagnosis within a few days	Unstable; cannot be positioned (orthopaedic injury); or diagnosis > 1 week
Difficult: pelvis usually not dilated	Authors: repair about 2–3 months after drainage ; scar makes surgery harder

- **Repair sequence:** good exposure → debride non-viable ureter and pelvis → **spatulated, tension-free ureteropelvic anastomosis** (mobilise kidney downward if needed) → over a **double-J stent for 4–6 weeks** → **retroperitoneal drain for several days** → nephrostomy optional (surgeon preference or tenuous repair).
- Delayed repair follows the same principles. **Always mention possible nephrectomy.**

Renal Pelvis Rupture

- **Rare;** possibly commoner with **pre-existing hydronephrosis or UPJ obstruction** (uncertain given rarity).
- Managed successfully with an **indwelling ureteral stent**, or by operative repair; with pre-existing UPJ obstruction, **delayed pyeloplasty after the leak heals.**
- **Fig. 62.8** Rupture of a dilated pelvis → stent + urinoma drain → no extravasation at stent removal → **delayed robotic-assisted laparoscopic pyeloplasty.**

KEY POINTS: OPERATIVE INTERVENTION INCLUDING URETEROPELVIC JUNCTION DISRUPTION AND RENAL PELVIS RUPTURE

- **Operate for** renal-bleeding instability, pulsatile/expanding haematoma, failed embolisation.
- **Vascular control before or immediately after opening Gerota fascia;** attempt salvage.
- **UPJ disruption:** acute repair if stable, else nephrostomy ± drain, delayed repair.
- **Pelvis rupture:** operation, or endoscopic/percutaneous drainage.

Concept — Surgery is reserved for bleeding that threatens life or defeats embolisation; everything else about the operation — contralateral kidney check, early vascular control, bolstered renorrhaphy — serves the second aim of salvaging the kidney without spending the patient.

Congenital Anomalies and Renal Injuries

- Duplex kidneys, fusion anomalies, hydronephrosis, hydroureter and UPJ obstruction are occasionally found at injury. **Risk of injury is not thought higher with hydronephrosis or UPJ obstruction, but this is controversial.**

Ureteropelvic Junction Obstruction

- **Commonest injuries are still grade 1–2, but leaks need more aggressive management because the obstructed UPJ delays drainage.**
- A leak is most likely a **grade 4 injury through thinned parenchyma;** consider pelvic laceration or UPJ disruption. **No ureteric contrast on excretory phase → retrograde pyelography.**
- **Pelvic laceration with apparent UPJ obstruction → early ureteral stent ± temporary perinephric drain;** nephrostomy ± stent if the leak persists; exploration may be needed.

- **After recovery:**

- **Stent removal → pyeloplasty weeks later** (authors: **robotic-assisted pyeloplasty 8–12 weeks later**, with minimal periureteral inflammation and scarring), or
- **Pyeloplasty with the stent still in place** — avoids infection in a poorly draining kidney after stent removal.

Duplex Kidney

- **Complete duplication in about 1–2% of the population; partial duplication (bifid pelvis) is commoner.**
- **Trap: a stent in the upper-pole ureter will not help a lower-pole leak.**

Fusion Anomalies and Ectopic Kidney

- **Managed like normal kidneys**, but horseshoe and ectopic kidneys **more often have hydronephrosis and UPJ obstruction and anomalous vasculature**, complicating management.
- **Fig. 62.9** Horseshoe kidney, grade 4 injury to a chronically obstructed left moiety → **stent, then delayed pyeloplasty**.

Concept — An anomalous kidney is not clearly easier to injure, but it drains worse — so its leaks persist and need earlier diversion, and its anatomy (duplication, fusion, aberrant vessels) dictates where drainage must be placed.

Follow-Up and Long-Term Issues

- At recovery, address **follow-up imaging, residual function of the injured kidney, long-term risk of hypertension and chronic kidney disease (CKD), and activity.**

Late Repeat Imaging and Assessment of Renal Function

- **Grade 1–2: scarring negligible** — AUA does not recommend repeat imaging.
- **Grade 3: scarring in about 50–60%; grade 4–5: close to 100%**, with some decline in differential function.
- Options: **CT at 3 months** (some) or **ultrasonography** (others). **Dimercaptosuccinic acid (DMSA) or mercaptoacetyltriglycine (MAG3) renography** measures function but rarely changes management — usually reserved for suspected major loss of function or hypertension.
- **Authors (ALARA): ultrasonography 2–4 months after grade 3–5 injury; DMSA** discussed for suspected functional loss (especially with hypertension), a very abnormal ultrasound, or after angioembolisation.
- **EAU: individualised radiological follow-up.**

Hypertension

- **Hypertension developing or persisting 30 days after injury → consider trauma-related hypertension.**
- **Uncommon after grade 1–2.** Adults after grade ≥ 3: **about 5–15%.** Children: **rare** (a small percentage on systematic review); risk rises with **higher grade and renal arterial injury.**
- **Mechanisms:** parenchymal ischaemia → **renin**; **parenchymal compression → renin (Page kidney)**; arteriovenous fistula or pseudoaneurysm; significant scarring.
- **Pediatric Trauma Society: blood pressure checks at follow-up** — frequency and duration undefined.
- **Workup:** nephrology referral, ultrasonography, renal scan for function and scarring, **magnetic resonance imaging (MRI) or CT** if arteriovenous fistula is suspected.
- **Treatment: angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB)**, since the hypertension is renin-mediated. **Poorly functioning, scarred kidney after grade 4–5 injury or angioembolisation → consider nephrectomy with nephrology.**
- **Page kidney** — subcapsular collection compressing parenchyma — is rare; drainage, spontaneous resorption and nephrectomy have all resolved hypertension; long-term outcome unclear.

Chronic Flank Pain

- **Uncommon**; not addressed in paediatric series.
- **Exclude a renal cause** (stone, UPJ obstruction): urinalysis and culture, ultrasonography, CT, MRI, renography; consider musculoskeletal and other causes.
- **Pain management referral (medical treatment, nerve blocks) before nephrectomy**, even for a poorly functioning kidney — **pain may persist after nephrectomy**.

Chronic Kidney Disease

- A concern after **nephrectomy, bilateral injury, or near-total loss of one kidney**.
- **Hypertension and diabetes are the commonest causes of CKD in people with a solitary kidney** → promote healthy lifestyle; refer if at risk (for example obesity).
- **Yearly blood pressure for a solitary kidney or a history of grade 3–5 injury**; offer nephrology referral.
- **Loss of a kidney from trauma contributes to end-stage renal disease in about 0.5%.**
- **Donor solitary kidney**: possibly a slight rise in hypertension; effect on CKD unknown.

Follow-Up and Activities With a Solitary Kidney

- **Organised sport carries lower risk, and lower-grade injury, than motor vehicles or motorised sport** (all-terrain vehicles, dirt bikes); but **sport-related renal injury is more often isolated**.
- **American Academy of Pediatrics: children with a normal solitary kidney may play sport, with protective padding for contact sport** (hockey, football) — **no study proves padding prevents renal injury**.
- **Higher-risk activities: dirt-bike riding, cycling, all-terrain vehicles** → counsel about risk; protective equipment.
- Authors clear children with a normal solitary kidney for organised sport with padding for contact sport, and **remind families that contact sport carries a higher risk of head injury**.

KEY POINTS: FOLLOW-UP AND LONG-TERM ISSUES

- **Scarring: none grade 1–2, about half grade 3, all grade 4–5.**
- **Grade 3–5: ultrasound first, then CT, MRI or DMSA if needed.**
- **Hypertension rare, after grade 3–5; CKD/ESRD very rare, even after nephrectomy.**
- **Solitary kidney: sport allowed, padding for contact sport.**

*Concept — Late sequelae scale with grade: grade 1–2 need nothing, grade 3–5 need an ultrasound and lifelong blood pressure checks, because **scarring and renin-mediated hypertension — not renal failure — are the realistic long-term risks**.*

URETERAL TRAUMA

Epidemiology

- **Rare**: in adult series < 5% of **penetrating trauma**, very rare in blunt trauma.
- **Suspect with**: penetrating injury near, or with trajectory towards, the ureter; **multivisceral or vertebral injury; rapid deceleration**.
- **Associations**: penetrating → kidney, vascular, bowel and solid-organ injury; blunt → **pelvic fracture**. **Mortality about 10%**, driven by associated injury. **Renal injury in about 20%.**
- **Delayed or missed diagnosis** → **more nephrectomy and longer hospital stay**.
- **Iatrogenic**: open or laparoscopic rectal, colonic, urological or gynaecological surgery — **suture ligation, crush, partial or complete transection, devascularisation, electrocautery**; rarely ureteroscopy. Highest

incidence in adults after **gynaecological and colorectal surgery (about 1%)**; in children limited to case reports and small series.

Presentation

- **Blunt:** usually with other injuries, including high-velocity pelvic fractures. **Penetrating:** entry near the ureter or trajectory towards it.
- **Haematuria is absent in 25–50%.**
- **Penetrating ureteral injuries are mostly diagnosed intraoperatively — yet about 10% are missed at laparotomy.**
- **High-velocity gunshot wounds (> 610 m/s; military assault rifles, AR-15-style rifles):** tissue injury far beyond the tract; the projectile may tumble or change course. **Renal pelvis, ureter and bladder may look normal or merely contused at laparotomy, then necrose and leak days later → place a ureteral stent and drains around urinary structures if any leak is suspected.**
- **Iatrogenic injury: diagnosis delayed in 50–60% —** until urine in a surgical drain, urinary ascites, ileus, fever or sepsis.

Imaging Findings

- **Delayed-phase CT is the standard (AUA).**
- **Complete transection: no contrast distal to the injury, with extravasation. Partial injury: some distal contrast, with extravasation.**
- Confirmation or diagnosis also by **retrograde pyelography** or intravenous pyelography.

American Association for the Surgery of Trauma Classification

TABLE 62.4 URETERAL INJURY: AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SCALE

Grade	Injury	Management implied
I	Contusion or haematoma without devascularisation	May need no intervention
II	< 50% transection	At least endoscopic; possibly open or laparoscopic repair
III	≥ 50% transection	At least endoscopic; possibly open or laparoscopic repair
IV	Complete transection, < 2 cm devascularisation	Open or laparoscopic repair, immediate or delayed
V	Avulsion, > 2 cm devascularisation	Open or laparoscopic repair, immediate or delayed

Modified from Moore EE et al., Organ injury scaling III, J Trauma 1992.

- **Two thresholds carry the scale: 50% of circumference and 2 cm of devascularisation.** Grade correlates with complexity of repair and associated injury; **location of injury then decides the repair.**

Management of Traumatic Ureteral Injury

Immediate Diagnosis

- **Partial or complete transection found intraoperatively in a stable patient (penetrating, blunt or iatrogenic) → repair under the same anaesthetic (AUA, EAU).** Repair type depends on **length and location.**
- **Contusion:** ureteral stent, or excision and repair, according to the situation.
- **Unstable → damage control, delayed repair:** ligate the ureter + nephrostomy; bring the ureter to the skin; bring a ureteral stent out to the skin; or, if no time, **leave a large drain beside the injury.**

Delayed Diagnosis

- **Diagnosed within the first week: immediate repair is an option.**
- **Diagnosed after 5–7 days:** classically **nephrostomy ± drain or stent, with planned delayed repair** — optimal wait unknown; **6–12 weeks** commonly advised.
- **Alternative: attempt a ureteral stent across the injury** — partial and even complete transections have healed without open surgery (reported success widely variable, roughly 15–75%).
- **AUA: attempt stenting for delayed partial injuries** (reported for complete transections too). **EAU:** endourological management first for most delayed diagnoses; open repair for failures.

URETERAL INJURY: IMMEDIATE VERSUS DELAYED DIAGNOSIS

Immediate (intraoperative, or within 1 week)	Delayed (> 5–7 days)
Repair now if stable — same anaesthetic	Try a stent across the injury; if it fails, nephrostomy ± drain
Unstable → damage-control diversion	Delayed reconstruction at 6–12 weeks
Contusion → stent or excision and repair	EAU: endourology first, open repair for failure

Proximal or Mid-Ureteral Injury Surgical Repair

- **Very proximal injury: as for UPJ disruption** — spatulated, tension-free ureteropelvic anastomosis.
- **Partial transection, mid to upper ureter: ureteroureterostomy** (spatulated, tension-free) **or closure of the partial transection without narrowing the lumen.**
- **Transureteroureterostomy is less desirable in children** — long life expectancy, and it puts the other kidney at risk.
- **Boari flap can reach the mid to proximal ureter** when the distal ureter is unavailable.
- **Stent for 4–6 weeks** (optimum unknown). Paediatric outcomes favourable.

Distal Ureteral Injury Surgical Repair

- **Ureteroureterostomy, or ureteral reimplantation with or without psoas hitch, over a stent,** for most repairs; favourable paediatric outcomes.

Lengthy Ureteral Injuries

- **Very rare.** Options: **long Boari flap, ileal ureter, autotransplantation, reconfigured bowel segment (Yang–Monti principle), buccal mucosa ureteroplasty.** Highly individualised.
- **Nephrectomy** is an option, especially for a **poorly functioning kidney with a normal contralateral kidney.**

URETERAL REPAIR BY LOCATION AND LENGTH

Level	Repair
UPJ / very proximal	Spatulated tension-free ureteropelvic anastomosis
Mid to upper	Ureteroureterostomy or primary closure of partial tear; Boari flap if distal ureter unusable; avoid transureteroureterostomy in children
Distal	Ureteroureterostomy or reimplantation ± psoas hitch
Long segment	Long Boari flap, ileal ureter, Yang–Monti, buccal mucosa, autotransplantation; nephrectomy if poor function and normal contralateral kidney

Follow-Up Imaging

- **Recommended after any ureteral injury.**
- **Sequence:** stent removed → **ultrasound at 4–6 weeks** → if no hydronephrosis, **at least yearly ultrasound for 2–3 years** → concerning hydronephrosis ± flank pain → **MAG3 diuretic renogram** (function and drainage) → suspected stricture → **antegrade and retrograde pyelography** for length and location.

- **Post-injury stricture:** endoscopic dilation or incision is advocated by some, **but a significant minority recur.**

KEY POINTS: URETERAL INJURIES

- Rare; **mostly penetrating**; often with other injuries; **missed at laparotomy.**
- **Diagnose by delayed-phase CT or retrograde pyelography.**
- **< 1 week: immediate repair if stable; > 1 week: try a stent, often nephrostomy and delayed repair.**
- Children repaired as adults.

Concept — Ureteral injury is a disease of missed diagnosis: haematuria is often absent and laparotomy misses some, so suspicion comes from mechanism and the delayed-phase CT — **and the timing of diagnosis, more than the injury itself, decides between immediate repair and diversion with delayed reconstruction.**

BLADDER INJURIES

Presentation

- Penetrating bladder injury occurs but is **less common than in adults**; blunt trauma causes more than 80% of paediatric bladder and urethral injury.
- **Isolated bladder trauma is rare in children but commoner than in adults** — possibly because **more of the child's bladder is intraperitoneal.**
- Severity scoring mirrors adults (Table 62.5); **the critical distinction is intraperitoneal versus extraperitoneal.**

TABLE 62.5 BLADDER INJURY: AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SCALE

Grade	Injury
I	Contusion, intramural haematoma, or partial-thickness laceration
II	Extraperitoneal wall laceration < 2 cm
III	Extraperitoneal ≥ 2 cm, or intraperitoneal < 2 cm
IV	Intraperitoneal laceration ≥ 2 cm
V	Intra- or extraperitoneal laceration extending into bladder neck or ureteric orifice (trigone)

Modified from Moore EE et al., Organ injury scaling III, J Trauma 1992.

- **Pattern:** at equal size, **an intraperitoneal tear scores one grade higher; trigone or bladder-neck extension is grade V regardless.** Trap: a small intraperitoneal tear is only grade III, yet still needs repair.

Pelvic Fracture

- In national registry data, **most children with bladder trauma also had a pelvic fracture**, the likely cause. Injury is found through **haematuria or a mechanism prompting imaging.**
- **Children have pelvic fractures about half as often as adults** — the pliable pelvis needs more force to fracture — so fracture has been hypothesised to signal more severe intra-abdominal injury, though this is not consistently borne out.
- **Only about half of children with a bladder injury have a pelvic fracture** — **the bladder is at risk without bony injury.**
- Pelvic injury patterns in small children differ from adolescents and adults, changing in the **early teenage years.**

CHILD VERSUS ADULT: LOWER URINARY TRACT TRAUMA

Feature	Child	Adult
Bladder position	More intraperitoneal → isolated bladder injury commoner	Pelvic organ
Pelvic fracture	Half as frequent; needs higher energy → more concomitant urethral injury	Commoner
Bladder injury without fracture	About half	Less common
Level of posterior urethral injury	Supraprostatic and prostatic urethra	Bulbomembranous junction
Bladder neck injury	Commoner	Less common
Complications after urethroplasty	Same types, lower rates	Higher rates

Imaging

- **Paramount triggers: gross haematuria or urinary retention.**
- **AUA and EAU: cystography (plain or CT) for pelvic fracture with gross haematuria; gross haematuria with a mechanism concerning for bladder injury; pelvic fracture with clinical signs of bladder rupture.**
- **Delayed-phase CT may miss bladder injury → retrograde contrast study (CT or conventional cystography).**
- **Haematuria with pelvic fracture: about half have bladder rupture.**
- Cystography via an existing catheter **avoids IV contrast; retrograde urethrography (RUG)** can be done at the same time.

Management

- **Location determines management**, as in adults: intraperitoneal, extraperitoneal, bladder neck.

EXTRAPERITONEAL VERSUS INTRAPERITONEAL BLADDER RUPTURE

	Extraperitoneal	Intraperitoneal
Default	Catheter drainage, no repair in most	Repair as a rule (AUA, EAU: absolute indication)
Why	Urine contained	Uncontained peritoneal urinoma — morbidity and mortality
Repair indicated when	Bone fragments exposed in bladder; pelvic internal-fixation hardware; rectal or vaginal laceration; abdominal surgery needed; bladder-neck involvement	Always (non-operative management in selected cases is controversial and needs continuous bladder and urinoma drainage)
How	Closed at time of other surgery (e.g. pelvic fixation), midline incision; two-layer running absorbable ; ± perivesical drain	Laparotomy, with or without other exploration
Catheter	1–2 weeks → cystogram at removal	About 1–2 weeks → cystogram at removal

- **Why bone and hardware matter:** exposed fragments risk complex pelvic bone infection; hardware may be seeded by infected urine.
- **Bladder neck injury — commoner in children and difficult:** a tear through the neck needs **formal repair**. **Complete avulsion → poorer continence, especially in girls**, who rely more on the bladder-neck

mechanism. **Continence is notoriously poor** — many need secondary procedures, even **bladder neck closure and diversion**.

Concept — Location, not size, decides bladder management: *extraperitoneal urine is contained and drains through a catheter, intraperitoneal urine is not — and any contact with bone, hardware, rectum, vagina or the bladder neck converts an observable injury into one that must be closed.*

Urethral Injuries

- Classified as posterior or anterior, with **adult AAST severity scoring** (Table 62.6); the **Goldman classification** is widely used for its clinical utility (Table 62.7).
- **Posterior injuries, including bladder-neck injuries, are caused by pelvic fracture; straddle injury damages the anterior (bulbar) urethra, as in adults.**
- **Pelvic fracture in a child implies higher energy, so concomitant urethral injury is substantially more likely.**
- Anterior and pendulous urethral injuries: see Penile Injury.

TABLE 62.6 URETHRAL INJURY: AAST ORGAN INJURY SCALE, AND TABLE 62.7 GOLDMAN CLASSIFICATION FOR BLUNT URETHRAL TRAUMA

Grade / type	AAST OIS (Table 62.6)	Goldman (Table 62.7)
I	Contusion: blood at meatus, normal RUG	Posterior urethra stretched but intact
II	Stretch: urethral elongation, no extravasation	Rupture above an intact urogenital diaphragm
III	Partial disruption: extravasation, contrast reaches bladder	Rupture through the urogenital diaphragm, involving proximal bulbar urethra
IV	Complete disruption: extravasation, no bladder filling, < 2 cm separation	Extraperitoneal bladder rupture extending into proximal urethra
V	Complete transection, ≥ 2 cm separation, or extension into prostate or vagina	Anterior urethral injury

Modified from Moore EE et al., J Trauma 1992; Goldman SM et al., J Urol 1997.

- **The two systems ask different questions: AAST grades severity by contrast behaviour and gap length; Goldman classifies anatomical level relative to the urogenital diaphragm.** Key AAST threshold: **2 cm of separation**; key AAST discriminator between partial and complete disruption: **whether contrast reaches the bladder**. Goldman type V is the only anterior injury.

Evaluation

- **Urethrography when the mechanism suggests pelvic trauma** (straddle injury, motor vehicle collision with pelvic fracture).
- **History:** urinary retention, gross haematuria. **Examination:** **blood at the meatus or from the vagina**, perineal haematoma → radiographic and/or cystoscopic assessment.
- Existing CT or radiographs help confirm pelvic bony or organ injury.
- **AUA: RUG in every case of blood at the meatus with pelvic trauma.**

Management

- **Catheter not yet placed → RUG before any attempt** — harder in awake children but critical to distinguish partial from complete disruption.
- **Catheter in → RUG around it, or voiding cystourethrography (VCUG); cystourethroscopy under anaesthesia if inconclusive.**

- **A catheter that passed suggests the injury is not complete** → confirm on RUG → **catheter drainage and reassess in several weeks**. Resultant stricture can be treated by urethrotomy, but **about half recur** — avoid repeated endoscopic attempts.
- **Urethral drainage not achievable** → **suprapubic tube is mandatory**.

POSTERIOR URETHRAL DISRUPTION: INITIAL OPTIONS

Option	For	Against	Verdict
Endoscopic primary realignment (cystoscopic + fluoroscopic)	Some good paediatric results in complete disruption or avulsion	High stricture rates; bladder neck hard to find; irrigation expands the pelvic haematoma and bladder–floor gap; may compromise later open repair	Stable patient: a brief attempt to pass a catheter; no prolonged attempts
Open primary realignment / immediate urethroplasty	Some report results similar to delayed repair	Often needs further procedures; releases tamponade → deterioration ; poor view → over-debridement → more erectile dysfunction and incontinence	EAU advises against immediate urethroplasty
Suprapubic diversion, delayed repair	Scar matures; haematoma resolves; bladder descends; gap shrinks	Delay of months	Standard
Exception: associated bladder neck injury	Prevents urinoma and infection	—	Repair immediately

- **Delayed repair:** wait **several months**; **re-image near the repair date** (cystography + RUG, cystoscopy from below and via the suprapubic tract) to measure the gap.
- **Fig. 62.10** Girl of 6 years: **bladder elevated off the shattered pelvis**, then **descended after haematoma resolution**, before reconstruction.
- **In children, excision and primary anastomosis is usually achievable without tension, even in severe injury**; in a large series the transperineal route sufficed in the great majority, with an abdominal approach needed for a small minority.
- **Technique:** lithotomy, abdomen prepared → **midline perineal incision** → mobilise urethra (to glans if needed) → transect through scar → **sound via suprapubic tract into bladder neck, incise onto it** → **excise scar to well-vascularised tissue** → spatulate → **interrupted absorbable anastomosis over Foley** → tension: **partial pubectomy, then bladder mobilisation from above** → catheter **2–3 weeks** → suprapubic tube out after **voiding trial and cystogram**.
- **Posterior sagittal transanorectal approach** is an alternative.
- **Gap too long:** skin or buccal inlay grafts or flaps; interposition of appendix, bowel or ureter. **Defects < 5 cm are usually amenable to excision and primary anastomosis** — almost all initial repairs; reoperations may be longer.
- **Outcomes favourable.** Complications as in adults but at **lower rates**: retention (atony or recurrent stricture), incontinence, penile shortening, erectile dysfunction. **AUA: follow up for at least 1 year.**
- **Recurrent stricture in about 10%** → internal urethrotomy (**high failure; better for strictures < 1 cm**) → **redo urethroplasty**, itself more prone to recurrence.
- **Incontinence in up to 20%**, linked to **proximal injury (bladder neck, prostatic urethra)**; with **bladder-neck extension continence is poor regardless of timing or type of repair**.
- **Erectile dysfunction in up to half; correlates with preoperative dysfunction (with potential long-term recovery)**, fracture severity and defect length; associated with **lateral prostatic displacement and gap > 2.5 cm; commoner after acute open realignment** — a reason to avoid immediate realignment.

Urethral Disruption in Females

- **Rare; almost exclusively with pelvic fracture; usually a complete disruption below the bladder neck. About 5% of female pelvic fractures.**
- **Suspect with haematuria, retention, or vaginal or rectal injury.** Causes: crush, motor vehicle collision, falls with pelvic fracture.
- **Suprapubic drainage → delayed repair several months later. External sphincter usually preserved → excision and primary anastomosis, lithotomy, abdomen prepared for bladder mobilisation.**
- **Continence favourable when the bladder neck is spared;** incontinence, recurrent stricture and fistula occur.

KEY POINTS: BLADDER AND URETHRAL INJURY

- **Pelvic trauma + gross haematuria or retention + fracture → RUG and/or cystography.**
- **Extraperitoneal repair:** hardware, bone fragments, bladder neck, abdominal surgery. **Intraperitoneal:** repair.

Concept — Posterior urethral disruption follows "drain now, reconstruct later": suprapubic diversion lets the haematoma resolve and the bladder descend, so a daunting acute gap becomes a short scar suited to excision and primary anastomosis — **the bladder neck is the one structure repaired at once.**

GENITAL INJURY IN THE PAEDIATRIC POPULATION

- **Most genital injuries are accidental, but abuse must always be assessed** — penile or scrotal trauma in boys, vaginal trauma in girls.
- **About one in seven boys and one in three girls are sexually abused in childhood, mostly penetratively.** Signs are not always obvious; **involve social services** to establish the safety of the home before discharge.
- **When abuse is suspected, exclude concomitant injury — especially to the anus and rectum, which often goes unnoticed.**

Penile Injury

- Causes: **circumcision (the majority)**, hair tourniquet, motor vehicle collisions, animal bites, zippers, burns and scalds. Severity and type direct management.

TABLE 62.8 PENILE INJURY: AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SCALE

Grade	Injury
I	Cutaneous laceration or contusion
II	Buck fascia (cavernosum) laceration without tissue loss
III	Cutaneous avulsion; laceration through glans or meatus; cavernosal or urethral defect < 2 cm
IV	Partial penectomy; cavernosal or urethral defect ≥ 2 cm
V	Total penectomy

AAST penile organ injury scale (Moore EE et al.).

- **The 2-cm cavernosal or urethral defect separates grade III from IV.**
- **Circumcision injury:** complication rates vary with **technique and training**; severe injury (partial or complete glans amputation) mostly with **inadequately trained operators**; **less common in hospital and clinic settings**; **commoner with the Mogen device** — **no protective bell over the glans**. Careful lysis of preputial adhesions probably prevents most complications, even with a Mogen clamp.

- **Excess skin removal:** observation with **healing by secondary intention** usually gives good cosmesis; severe cases may need **tissue expanders and full-thickness skin grafts**.
- **Hair or thread tourniquet:** partial or circumferential injury → **urethral injury or even penile loss**; slow onset, often painless, the hair may be too fine to see → **late recognition**.
- **Amputation (circumcision or strangulation):** careful repair gives good cosmesis with or without **microvascular reanastomosis**; catheter 1–2 weeks after urethral anastomosis. Tourniquet injury is often less severe and conservative; surgical outcomes favourable. **Complete loss or failed reanastomosis → neophalloplasty** (penile agenesis techniques): **good cosmesis, mixed function**.

PENILE URETHROCUTANEOUS FISTULA: HAIR TOURNIQUET VERSUS CIRCUMCISION

Hair tourniquet	Circumcision
Proximal shaft	Subcoronal
Catheter drainage first → multilayer closure if unhealed	May need glans splitting and repair as for hypospadias
Favourable; occasional stricture or recurrence	—

- **Toilet-seat crush:** common in toilet-training boys; **mostly minor, conservative, outpatient**.
- **Burns:** rare, but **commoner in children than adults**; **mostly scalds**, rarely isolated to the genitalia, may cause major skin loss → graft or flap. **Harvest from sites expected to be less hairy after puberty**; **full-thickness, unmeshed grafts for the shaft** (natural appearance, moves freely over the corpora); **bolster 5–7 days**, limited activity.
- **Zipper entrapment** — among the commonest penile injuries: minor but awkward. In a randomised animal-model trial, **mineral oil manipulation had the highest success**; **cutting the zipper teeth with trauma shears caused least skin trauma and was preferred** over removing the median bar.
- **Animal bites:** debridement and copious irrigation; if not infected wounds may be closed, but **usually best left to heal by secondary intention**.

Scrotal Injury

Epidemiology

- **Testicular trauma is rare:** < 5% of operations for acute scrotum.
- **Penetrating:** superficial (skin, dartos) or deep (into tunica vaginalis). **Blunt:** from minor haematoma to rupture with extruded seminiferous tubules.

TABLE 62.9 SCROTAL INJURY AND TABLE 62.10 TESTICULAR INJURY: AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SCALES

Grade	Scrotum (Table 62.9)	Testis (Table 62.10)
I	Contusion	Contusion or haematoma
II	Laceration < 25% of scrotal diameter	Subclinical laceration of tunica albuginea
III	Laceration ≥ 25% of scrotal diameter	Tunica albuginea laceration, < 50% parenchymal loss
IV	Avulsion < 50%	Major tunica albuginea laceration, ≥ 50% parenchymal loss
V	Avulsion ≥ 50%	Total destruction or avulsion

AAST scrotal and testicular organ injury scales (Moore EE et al.).

- **Thresholds:** scrotum 25% (laceration) and 50% (avulsion); testis 50% parenchymal loss.

Imaging

- As with suspected torsion, a convincing history or examination justifies immediate exploration without imaging.

- **Ultrasonography helps detect rupture: discontinuity of the tunica albuginea, haematocoele, loss of Doppler flow to part or all of the testis.**

Management

- **Penetrating, tunica vaginalis intact → washout and closure** — often in the emergency department under local anaesthetic with minimal or no sedation.
- **Tunica vaginalis breached, or testis extruded → formal debridement and washout in theatre;** drain for several days if needed.
- **Blunt:** suspected rupture (tunica albuginea not intact, or haematocoele) → **exploration. Confirmed rupture → acute debridement and tunica albuginea repair** (AUA, EAU). Some advocate conservative management even with rupture — **atrophy under 20%** in one series.
- **Scrotal skin loss:** local **advancement flaps**, even after extensive debridement; otherwise **split-thickness meshed graft** from abdomen or thigh.
- **Follow-up at about 6 weeks: examination and ultrasonography** — confirm resolution and **exclude testicular pathology masked by haematoma**, particularly after non-operative management.

GENITAL SKIN COVER: PENIS VERSUS SCROTUM

Penile shaft	Scrotum
Full-thickness, unmeshed graft from sites less hairy after puberty	Local advancement flap first; else split-thickness meshed graft (abdomen, thigh)
Bolster 5–7 days	—

Vaginal Injury

- **Uncommon;** two categories — **straddle** and **penetrative**. AAST vaginal and vulvar scales exist (Tables 62.11, 62.12) but are **rarely cited**; no alternative is widely used.

TABLE 62.11 VAGINAL INJURY AND TABLE 62.12 VULVAR INJURY: AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA ORGAN INJURY SCALES

Grade	Vagina (Table 62.11)	Vulva (Table 62.12)
I	Contusion or haematoma	Contusion or haematoma
II	Superficial laceration (mucosa only)	Superficial laceration (skin only)
III	Deep laceration (into fat or muscle)	Deep laceration (into fat or muscle)
IV	Complex laceration into cervix or peritoneum	Avulsion of skin, fat or muscle
V	Injury into adjacent organs (anus, rectum, urethra, bladder)	Injury into adjacent organs (anus, rectum, urethra, bladder)

AAST vaginal and vulvar organ injury scales (Moore EE et al.).

- **The two scales are identical except grades II and IV** — mucosa versus skin, and cervix/peritoneum versus avulsion.
- **Straddle injury is by far the commonest (about 80%); penetrating about 5%.**
- **Straddle injuries do not ordinarily involve the perianal region, hymen or vagina** — such injuries suggest **penetration**, intentional or not.
- **Any history or sign of penetrating vaginal injury → suspect abuse.** Signs: **injury to neighbouring organs (bladder, urethra, anus, rectum), internal vaginal injury, hymenal injury.** Hymenal injury correlates strongly with sexual abuse and suggests it **unless the injuring event was witnessed as accidental.**
- **Vaginal injury may accompany pelvic fracture** with urethral, bladder and rectal injury.

- Series bottom line: **bleeding the commonest complaint, straddle the commonest mechanism, labia mostly involved, urethra spared**; ED sedation often sufficed, **theatre rarely** — more likely with uncontrolled bleeding, younger age, difficult examination or another procedure. **Examination under anaesthesia or sedation allows cystoscopy, vaginoscopy and anoscopy (general surgeon).**

Management

- **Straddle injuries usually cause minor labia minora trauma, heal well without scarring, and rarely need theatre.**
- **Older girls with larger injuries — especially involving the hymen, or with perineal bleeding, haematoma or swelling — more often need operative repair.**
- **External appearance may not reflect the severity of injury, or whether any trauma occurred.**
- In prospective follow-up of anogenital trauma (more than half from abuse among the girls), **few had significant anatomical evidence of injury later — most heal without obvious scar**; repair of the posterior fourchette acutely is recommended to improve healing of the introitus.
- **Urethral, bladder, anal and rectal injuries can be missed → examination under anaesthesia if abuse is strongly suspected.**
- **Healing:** superficial injuries (abrasions, contusions, superficial lacerations) heal rapidly; deep lacerations take several weeks; **hymenal injuries heal rapidly, mostly without scar. Therefore a normal examination does not exclude sexual abuse.**

KEY POINTS: GENITAL INJURY

- **Suspect abuse; exclude anorectal injury.**
- **Reduce adhesions → avoid most circumcision injury.**
- **Ultrasound rupture → explore.**
- **Straddle = most vaginal injury; deeper injury suggests penetration → abuse evaluation.**

Concept — In genital trauma the mechanism is itself a diagnosis to be made: straddle injuries stay external, so any injury reaching the hymen, vagina, anus or adjacent organs implies penetration and obliges an abuse assessment — while rapid scarless healing means a normal examination never excludes abuse.